🌿 ESP32 Air Quality & Environmental Analytics

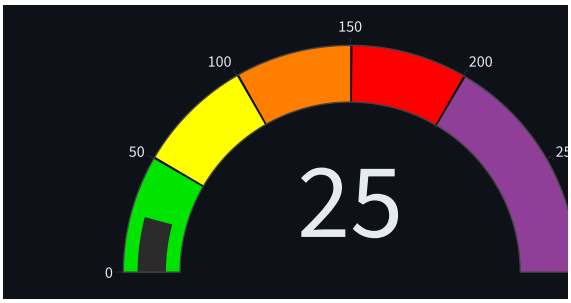
🟢 Connected

2026-10-12 17:23:11

📊 Core Metrics

🌡️ Temperature	💧 Humidity	📊 Air Quality Index	⚡ ADC Voltage	💨 Est. CO_2 / VOC
33.8 °C	34.0 %	25	2.33 V	1301 ppm
↓ -0.0 °C HeatIdx	↑ 15.8 °C Dew Pt			

AQI Status



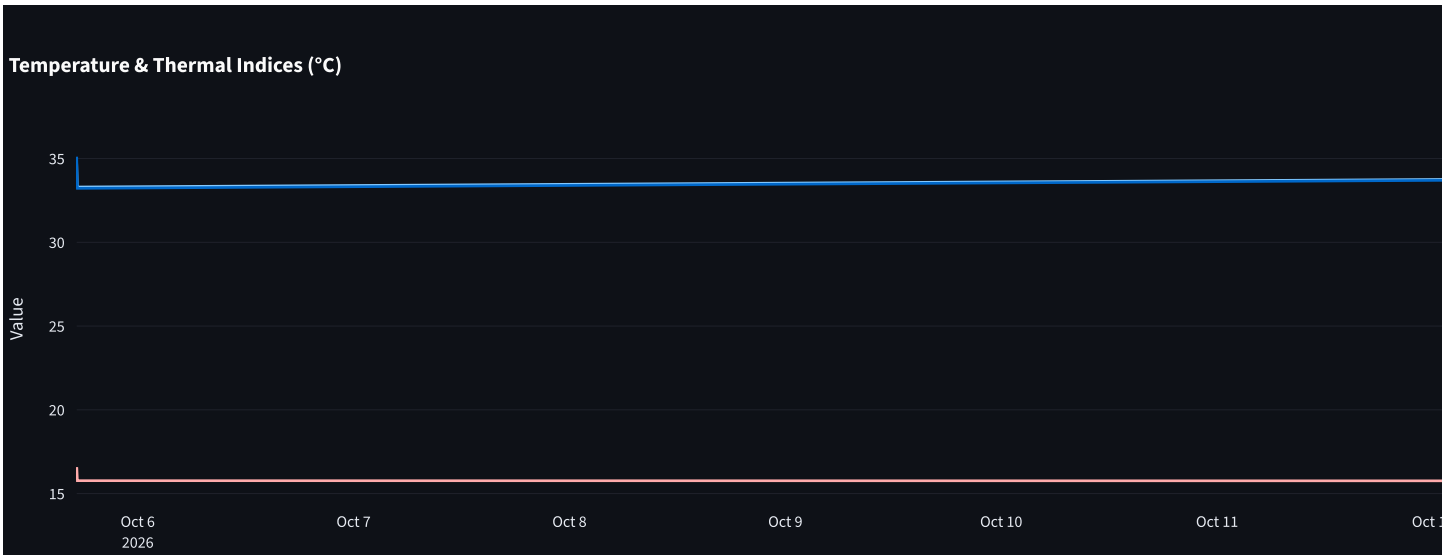
Classification: Good | **Comfort Status:** Heat Stress Risk

🔬 Atmospheric Calculations

Vapor Pressure Deficit (VPD)	Gas Sensor Ratio (Rs/R0)
3.472 kPa	0.42
Absolute Humidity	**Raw MQ-135 Counts**
12.64 g/m³	2887

📈 Historical Trends

Temperature & Humidity | Air Quality & Gas | VPD & Dew Point



timestamp

📚 Beginner's Guide to Understanding Dashboard Parameters

New to environmental monitoring? Here is a simple explanation of what each parameter measures and why it matters.

🌡️ Thermal & Moisture Parameters

- **Temperature (°C)**
  What it is: Standard measurement of how warm or cold the air is.
  Ideal Range: **20°C – 26°C** for indoor room comfort.

- **Relative Humidity (%)**
  What it is: The percentage of water vapor currently in the air compared to the maximum amount the air can hold at that temperature.
  Ideal Range: **30% – 60%**. High humidity promotes mold growth, while low humidity causes dry skin and throat irritation.

- **Heat Index (°C)**
  What it is: The "feels-like" temperature. High humidity prevents sweat from evaporating, making warm air feel even hotter to the human body.

- **Dew Point (°C)**
  What it is: The exact temperature air needs to cool down to for water droplet condensation (dew/fog) to form. If Dew Point is close to Temperature, the air feels muggy and sticky.

- **Absolute Humidity (g/m^3)**
  What it is: The exact weight of water vapor contained in one cubic meter of air, regardless of temperature.

💨 Air Quality & Sensor Engineering Metrics

- **AQI (Air Quality Index)**
  What it is: A standard 0–500 scale that tells you how clean or polluted the surrounding air is.
  Good Range: **0 – 50** (Safe) | **>100** (Unhealthy for sensitive groups).

- **Vapor Pressure Deficit (VPD in kPa)**
  What it is: Measures how "dry" the air feels to plants and lungs. It is the gap between maximum saturation pressure and current moisture pressure.
  Ideal Range: **0.8 – 1.2 kPa**. Very useful in greenhouses and plant growth rooms.

- **Est. CO_2 / VOC (ppm)**
  What it is: An estimated count of Parts-Per-Million (ppm) of carbon dioxide and Volatile Organic Compounds derived from the MQ-135 gas sensor.
  Ideal Range: **400 – 1000 ppm**. Above 1000 ppm usually indicates bad ventilation.

- **ADC Voltage (V) & Raw Counts**
  What it is: The direct electrical reading on the ESP32 chip pin (0 to 4095 raw value mapped to 0.0V – 3.3V). Higher values mean higher gas/smoke concentrations detected.